

Rethinking Synthetic Tabular Data Generation as an End-to-End Optimization Problem

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Abstract—Scientific discovery increasingly relies on high-quality, AI-ready datasets; however, data scarcity remains a persistent challenge across many scientific domains. This work introduces a two-stage framework for synthetic tabular data generation that integrates diverse generative models with automated post-generation curation to address scarcity in privacy-sensitive domains such as medicine and the social sciences.

Through extensive empirical evaluation across multiple model classes and scales, we show that large-scale models consistently outperform classical generative approaches in terms of statistical fidelity and downstream task performance. The proposed curation pipeline systematically improves data quality across a broad range of statistical and task-oriented metrics while reducing the presence of poorly reproduced samples and mitigating potential information leakage.

We further investigate the role of synthetic data augmentation within this framework. While naive augmentation provides inconsistent gains, controlled augmentation combined with model-aware curation preserves stable performance across selected model types and improves the overall reliability of synthetic datasets. These results establish the framework as a practical approach for generating high-quality, privacy-conscious synthetic tabular data applicable across a wide range of scientific tasks.

Index Terms—synthetic data, large language models, data curation, data augmentation, AI-ready datasets, tabular data, machine learning

I. INTRODUCTION

Synthetic tabular data generation represents a fundamental challenge in machine learning, with applications spanning privacy-preserving data sharing, data augmentation, and statistical modeling. Contemporary approaches rely predominantly on specialized generative models, including Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and diffusion models. However, these methods exhibit critical limitations when modeling complex multidimensional distributions with intricate dependencies.

Scientific discovery is entering a new era powered by artificial intelligence and machine learning, yet one major bottleneck remains: the lack of high-quality, domain-specific datasets that are truly AI-ready. In physics, chemistry, biology, and materials science, researchers face critical data scarcity that limits the application of modern ML techniques to accelerate discovery processes. Traditional experimental data collection is expensive, time-consuming, and often limited by ethical or safety constraints.

Recent advances in large language models (LLMs) have demonstrated remarkable capabilities across diverse domains, yet their potential for creating AI-ready scientific datasets tabular data synthesis remains largely unexplored in scientific contexts. Previous investigations have been constrained to demonstrating LLM generation quality of tabular data without addressing the critical gap between synthetic data creation and practical machine learning utility models with focus on small LLM (with fewer than 1 billion parameters - e.g. GREAT [1], PAFT [2]), that limiting our understanding of scaling and emergent capabilities of LLM in tabular synthesis domain.

The creation of AI-ready scientific datasets requires not only synthetic data generation but also intelligent curation processes that ensure statistical consistency, eliminate artifacts, and optimize data utility for downstream applications. Unlike raw synthetic data, AI-ready datasets must undergo systematic quality assessment, refinement, and validation to ensure reliable performance in scientific machine learning applications.

Our primary contributions include: (1) a novel automated curation pipeline that systematically balances privacy-utility trade-offs through multi-objective optimization, improving data quality while reducing privacy risks, (2) comprehensive evaluation revealing that large-scale LLMs outperform specialized models in generation quality but create inherent privacy vulnerabilities that curation only partially mitigates, (3) systematic assessment demonstrating expected quality degradation in synthetic data augmentation with promising exceptions—fine-tuned LLaMA showing superior stability for classification tasks and TabPFN demonstrating exceptional robustness to synthetic artifacts compared to traditional architectures, (4) evidence that LLM fine-tuning on synthetic data enables substantial privacy preservation while causing significant performance challenges, establishing this approach as primarily suitable for privacy-sensitive scientific domains rather than general-purpose enhancement of machine learning capabilities.

II. RELATED WORK

A. Synthetic Tabular Data Generation

Recent years have witnessed significant developments in synthetic tabular data generation methodologies. Bayesian

methods employ probabilistic models that ensure high interpretability but face substantial limitations when dealing with high-dimensional data or complex nonlinear dependencies. Generative Adversarial Networks (GANs) have introduced specialized architectures including CTGAN [3], PATEGAN [4], ADSPAN [5], and GANBLR [6] that address various aspects of tabular generation challenges. However, these GAN-based approaches frequently encounter training instability and mode collapse issues, limiting their reliability in production environments. To mitigate some of these issues, models like VECTGAN [7] have been proposed, which leverage vectorized embeddings to better capture complex feature interactions and improve training stability in tabular data synthesis.

Variational Autoencoders (VAEs) represent an alternative class of generative models with specialized versions for tabular data, including TVAE [8] and RTVAE [9]. Recent diffusion models, particularly TabDDPM [10], demonstrate impressive capabilities in modeling complex distributions through iterative denoising processes but require substantial computational resources.

The emergence of Large Language Models (LLMs) has introduced novel possibilities for tabular data synthesis. GREAT [1] pioneered this direction by adapting language models for tabular data through textual encoding. PAFT [2] advanced the field by introducing functional dependency awareness, while recent work [11] demonstrated LLM superiority over specialized models in preserving statistical properties. Additionally, GOGGLE [12] represents a novel approach that combines graph-based modeling with transformer architectures, enabling effective learning of relational dependencies in tabular data and enhancing generation quality.

B. Data Curation for AI-Ready Datasets

Data curation methodologies have emerged as critical components for creating reliable datasets for machine learning applications. Traditional approaches focus on manual quality assessment and expert-driven validation processes. However, the scale and complexity of modern scientific datasets necessitate automated approaches that can systematically identify and correct data quality issues. Data curation is a multifaceted process aimed at improving the overall quality and reliability of datasets. It can encompass data cleaning (filtering), imputation of missing values, and correction or refinement of labels (label enhancement). During the filtering stage, the primary objective is to eliminate outliers and anomalous values that exert a detrimental impact on data quality. The curation of synthetic tabular data introduces unique challenges. Unlike in the domain of synthetic data generation for question answering, where the validity of a sample can often be explicitly verified, tabular settings lack a definitive ground-truth criterion for determining whether a given instance is of high quality. Consequently, the evaluation of synthetic tabular data is inherently more complex. In practice, various heuristic metrics are employed for this purpose. For instance, model-driven evaluation can serve as a practical proxy for data curation quality, with popular choices including gradient-boosted decision trees (e.g.,

XGBoost) and deep neural networks (Seedat et al., 2023) [13]. Additionally, large language models (LLMs) have recently been explored as filtering mechanisms (Chen et al., 2023) [14], given their strong contextual reasoning abilities, which enable them to effectively discard low-quality samples. A further complication arises from the generation process itself. In particular, LLMs are prone to hallucinations, which may result in the creation of non-existent class labels or unrealistic values that do not align with the inherent data distribution. This motivates the need for systematic label enhancement. One promising strategy for mitigating annotation costs involves replacing manual labeling with auxiliary student models that support knowledge distillation and iterative label refinement (Saad-Falcon et al., 2023) [15]. The underlying assumption is that student models distilled from LLM teachers can outperform conventional weak supervision by producing more robust and reliable annotations. As LLMs continue to advance in capability and accessibility, the integration of such student models is poised to emerge as a cost-effective and scalable alternative to human annotation.

III. METHODOLOGY

A. Enhanced LLM-based Synthetic Data Generation

We formalize tabular data synthesis as the problem of generating synthetic dataset $\mathcal{D}_{syn} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ that preserves the multidimensional joint distribution $P(\mathbf{X}, Y)$ of the original dataset $\mathcal{D}_{real}$, where $\mathbf{X} \in \mathbb{R}^d$ represents feature vectors and Y represents target variables.

Based on observed scaling laws in language models, we implement differentiated approaches according to model size.

For models exceeding 10B parameters (Qwen-2.5 Coder-32B [16], GPT-OSS 20B [17]), we leverage emergent in-context learning capabilities without additional training. Data representation employs the "column_name is value" format for consistency with language modeling objectives. The prompt structure comprises: (1) direct generation instructions specifying row-by-row synthesis, (2) target row count specification, and (3) representative data exemplars in JSON format for enhanced structural clarity. Data preprocessing includes missing value removal, label encoding for categorical features, and four-decimal precision rounding for numerical values. Generation employs batch processing with 10 examples per batch to balance context utilization and computational efficiency, preventing context window overflow while maintaining sufficient pattern exposure.

The LLM evaluation spectrum spans two orders of magnitude: Qwen-2.5-Coder-14B and GPT-OSS 20B. This coverage enables comprehensive scaling analysis from sub-billion parameter models requiring intensive fine-tuning to large-scale models demonstrating strong few-shot capabilities. We evaluate LLM tabular generations against 3 specialized baseline models representing current state-of-the-art in tabular generation: GAN variants (CTGAN, DPGAN) and diffusion models (TabDDPM). All specialized models were tuned with Optuna [18] hyper-parameter optimization and the resulting

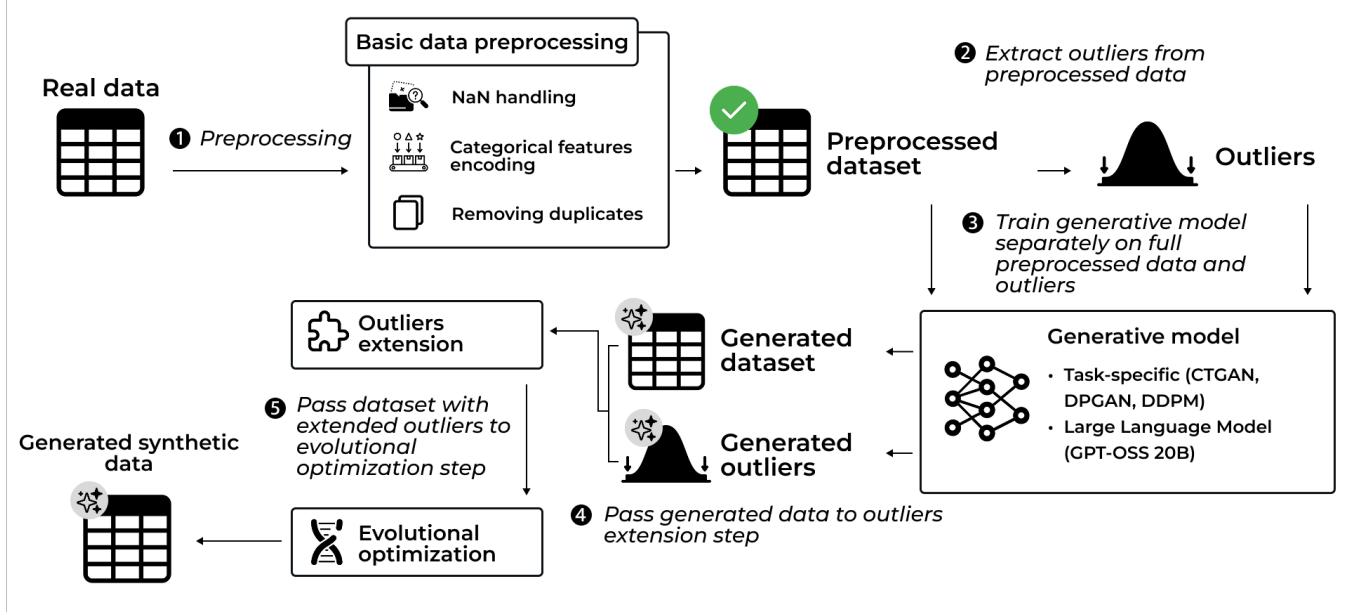


Figure 1. Overview of the full pipeline for synthetic tabular data generation.

configurations were documented in the supplementary repository accompanying this work.

B. Evolutionary Curation of Synthetic Data

Synthetic data curation can be formulated as the problem of identifying a subset of synthetic samples that optimizes a target performance criterion of downstream models. In this work, we focus on iterative identification and elimination of subsets of synthetic tabular data that negatively affect model performance on real-world data, thereby retaining only the most informative synthetic samples.

We formalize the curation of synthetic tabular data as a bilevel optimization problem. At the upper level, a subset of synthetic data is selected, while at the lower level a predictive model is trained on the selected subset. The objective is to maximize a target evaluation metric computed on a held-out real dataset. Formally, let D_s denote the synthetic dataset, D_r , the real dataset, and $\mathcal{H}$ a hypothesis class of models. The optimization problem can be written as:

$$\max_{D \subseteq D_s} \mathbb{E}_{f \sim \mathcal{H}} [\mathcal{M}(f(D), D_r)]$$

where $\mathcal{M}()$ denotes a chosen evaluation metric and the expectation is taken over models trained on the selected synthetic subset D .

Direct optimization of this objective is generally intractable due to the combinatorial nature of subset selection and the potentially large class of models and metrics of interest. To address this challenge and to allow for flexible incorporation of heterogeneous models and metrics, we resort to an evolutionary optimization strategy.

The optimization process is carried out using a population-based evolutionary algorithm. The initial population $\mathcal{P}^{(0)}$ is

constructed by randomly sampling subsets from the full synthetic dataset without replacement. Subsampling is performed uniformly at random, and subset sizes are drawn from a uniform distribution,

$$n_k \sim \mathcal{U}(n_{min}, n_{max}), n_k < N,$$

where $N = |D_s|$. This results in an initial population consisting of multiple candidate subsets of varying cardinalities, enabling exploration of both compact and more expressive synthetic representations.

Each individual in the population corresponds to a subset of synthetic rows and is evaluated via a fitness function. In our experiments, the fitness is defined as a convex combination of the Area Under the ROC Curve (AUC) achieved by two different model classes: a linear model and a gradient-boosted decision tree model (XGBoost). Specifically, models are trained exclusively on synthetic data subsets, while evaluation is performed on real data, ensuring that the fitness reflects the transferability of synthetic data to real-world distributions. While this work focuses on AUC-based fitness, the proposed framework is not limited to predictive performance metrics and naturally extends to alternative criteria, including statistical distances such as the Wasserstein distance between synthetic and real data distributions.

Selection is performed using tournament selection based on the fitness values, promoting individuals with superior generalization performance while maintaining population diversity.

To generate offspring, we employ a one-point crossover operator adapted to subset representations. Prior to crossover, for a given pair of parent subsets, we identify the elements that are unique to each parent, i.e., rows that do not appear in the other subset. The one-point crossover is then performed exclusively on these disjoint elements. This constraint guarantees that no

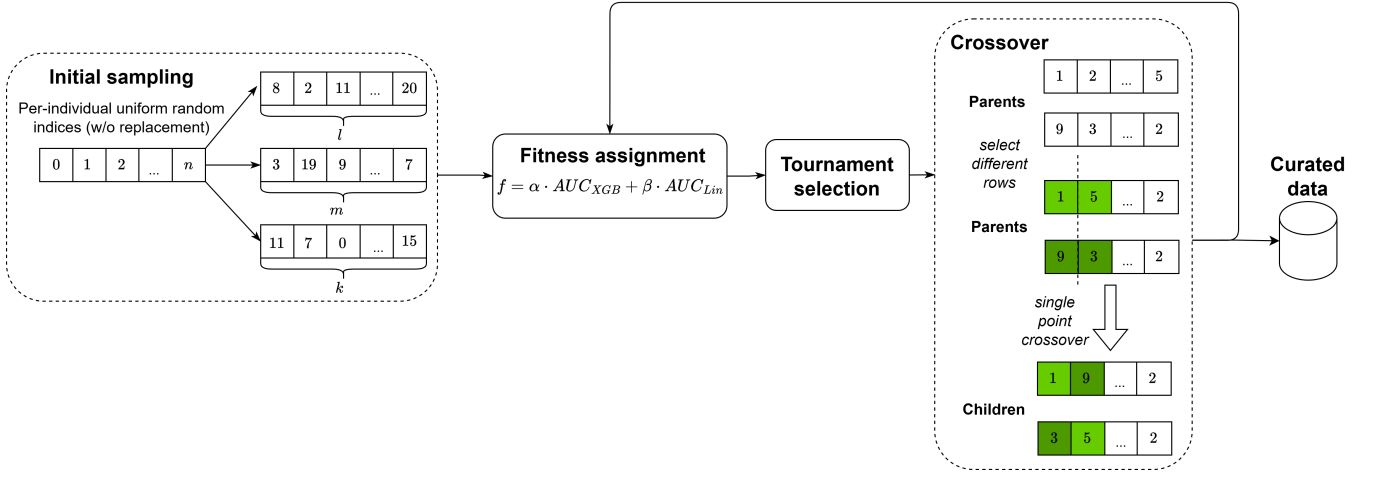


Figure 2. Schematic representation of the synthetic data curation procedure, showing iterative selection and refinement of data subsets to enhance predictive model outcomes.

Algorithm 1 Evolutionary Curation of Synthetic Tabular Data

Require: Synthetic dataset $\mathcal{D}_s$, real dataset $\mathcal{D}_r$, population size K , generations T , tournament size τ , subsample range $[n_{\min}, n_{\max}]$, fitness weight λ

- 1: **Initialize population** $\mathcal{P}^{(0)}$ by uniform random subsampling of $\mathcal{D}_s$ without replacement, with sizes $n_k \sim \mathcal{U}(n_{\min}, n_{\max})$
- 2: **for** $t = 0$ **to** $T - 1$ **do**
- 3: **Evaluate fitness** $\mathcal{F}(\mathbf{z}) = \lambda \cdot \text{AUC}_{\text{xgb}}(\mathbf{z}) + (1 - \lambda) \cdot \text{AUC}_{\text{lin}}(\mathbf{z})$ on $\mathcal{D}_r$ for all $\mathbf{z} \in \mathcal{P}^{(t)}$
- 4: **Generate offspring population** $\mathcal{P}^{(t+1)}$
- 5: **while** $|\mathcal{P}^{(t+1)}| < K$ **do**
- 6: Select parents $\mathbf{z}_a, \mathbf{z}_b$ via tournament selection
- 7: Extract unique subsets $\mathcal{U}_a = \mathcal{D}_{\mathbf{z}_a} \setminus \mathcal{D}_{\mathbf{z}_b}, \mathcal{U}_b = \mathcal{D}_{\mathbf{z}_b} \setminus \mathcal{D}_{\mathbf{z}_a}$
- 8: Perform deduplicated one-point crossover on $\mathcal{U}_a, \mathcal{U}_b$
- 9: Add offspring to $\mathcal{P}^{(t+1)}$
- 10: **end while**
- 11: **end for**
- 12: **return** $\mathcal{D}^* = \arg \max_{\mathbf{z} \in \mathcal{P}^{(T)}} \mathcal{F}(\mathbf{z})$

duplicate samples are introduced during recombination and preserves the set-based nature of the representation.

The evolutionary process is iterated until one of the following stopping conditions is met: (i) the improvement in the fitness function becomes negligible over successive generations, indicating convergence, or (ii) a predefined maximum number of generations T is reached.

C. Tail extension on synthetic data

IV. EXPERIMENTAL SETUP

A. Datasets

The experimental evaluation is conducted on five publicly available tabular datasets: Adult, Breast Cancer, Wine, US Location, and Diabetes. These datasets are commonly used

benchmarks in the literature and collectively cover a wide range of data characteristics, making them suitable for assessing the robustness of synthetic data curation methods.

The selected datasets differ substantially in both the number of samples and the number of features, as well as in feature semantics and statistical structure. This diversity allows us to evaluate the proposed method under varying levels of data sparsity, dimensionality, and sample complexity. In particular, the datasets span from low-dimensional settings with a limited number of numerical features (e.g., Wine) to higher-dimensional tabular data with heterogeneous feature types (e.g., Adult), and from relatively small datasets to datasets containing several thousand instances. Such variation is critical for analyzing the sensitivity of the curation algorithm to dataset scale and structural complexity.

All datasets correspond to supervised learning tasks and are treated as classification problems in our experiments. Synthetic data are generated for each dataset independently, and all downstream evaluations are performed on held-out real data drawn from the same source distribution.

Prior to model training and synthetic data generation, each dataset undergoes a standardized preprocessing pipeline. This includes handling missing values, feature normalization or scaling where appropriate, and basic data quality checks. The preprocessing steps are applied consistently across all datasets to ensure comparability of results and to isolate the effect of the proposed curation procedure from dataset-specific artifacts.

The choice of these datasets reflects a deliberate trade-off between benchmark diversity and experimental control. By limiting the evaluation to a compact yet heterogeneous set of datasets, we ensure reproducibility while enabling a systematic analysis of how dataset size and dimensionality influence the effectiveness of synthetic data curation.

Table I
SUMMARY OF DATASETS USED IN THE EXPERIMENTAL EVALUATION.

Dataset	Task Type	#Samples	#Features
Adult	Classification	48,842	14
Breast Cancer	Classification	569	30
Wine	Classification	178	13
US Location	Classification	5,000+	5
Diabetes	Regression	568	8

B. Compared Synthetic Generation Models and Evaluation Metrics

We evaluate against specialized baseline models representing current state-of-the-art in tabular generation: GAN variants (CTGAN, DPGAN), and diffusion models (TabDDPM). The LLM evaluation spectrum includes GPT-OSS 20B, Qwen2.5 Coder 14B.

Evaluation employs Synthcity [19] metrics suite across five categories: *sanity* (data validity checks), *statistical* (distributional fidelity measures), *performance* (downstream task utility), *detection* (synthetic data detectability), and *privacy* (privacy preservation assessment).

C. Implementation details

Task-specific generative models were trained on both CPU and GPU architectures. The GPU setup was a NVIDIA Tesla P100-SXM2 with 16 GB of memory, and the CPU setup was an Intel Xeon (Cascade Lake) 3.09 GHz 24-core processor with 24 GB of RAM.

Models trained on GPU included CTGAN, DDPM, DPGAN.

On average, the generation of 100 synthetic data samples using the few-shot paradigm with GPT-OSS 20B, Qwen2.5 Coder 14B required 3287 tokens, when each prompt included 10 real data examples. For comparison, the token consumption for generating 100 instances with prompts containing 1, 5, and 10 examples amounted to 14325, 5505, and 3287 tokens, respectively.

V. RESULTS AND ANALYSIS

A. Efficiency of data curation and tail extension algorithms

1) *Efficiency of data curation*: We compare the results obtained before and after applying the distribution tail amplification algorithm to raw synthetic data in order to demonstrate the potential of this approach for improving data quality. As shown in Table 2, tabular data curation leads to consistent improvements in downstream machine learning performance, as measured by the Linear syn-ID, Linear syn-OOD, XGBoost syn-ID, and XGBoost syn-OOD metrics.

The most pronounced gains are observed for data generated by DDPM, which can be attributed to the relatively low baseline performance prior to curation. This suggests that the proposed method is particularly effective in scenarios where the initial synthetic data quality is limited.

Importantly, the proposed post-processing procedure does not degrade statistical fidelity or privacy-related metrics. In several cases, we observe a slight improvement in privacy

metrics, which indicates that the tail amplification procedure does not exacerbate memorization effects and may even contribute to a mild regularization of sensitive regions of the data distribution. Overall, these results highlight the practical utility of the proposed method as a lightweight and model-agnostic enhancement step for synthetic tabular data.

2) *Tail extension*: We compare the results obtained before and after applying the distribution tail augmentation algorithm to the raw synthetic data. As shown in Table 3, the application of tail augmentation leads to a consistent improvement in statistical similarity metrics, including the Kolmogorov–Smirnov test statistic and the inverse Kullback–Leibler divergence. These improvements indicate a closer alignment between the augmented synthetic data distribution and the reference target distribution, particularly in low-density regions.

In addition, we observe moderate but systematic gains in downstream machine learning model performance metrics. Although the absolute improvements are limited, their consistency suggests that the original synthetic dataset lacked certain critical subpopulations that are essential for learning robust decision boundaries. The presence of such underrepresented regions disproportionately affects the generalization capability of machine learning models, especially in regimes where rare but informative samples play a significant role.

The observed behavior supports the hypothesis that tail augmentation effectively restores informative structure in the distribution extremes, thereby improving the representational completeness of the synthetic data. Importantly, similar to data curation procedures, tail augmentation has a negligible impact on metric categories that were not explicitly targeted during optimization. This indicates that the method introduces minimal unintended distortions and preserves the overall statistical integrity of the dataset.

These results demonstrate that distribution tail augmentation serves as a targeted and controlled mechanism for improving both statistical fidelity and downstream model performance, without inducing adverse effects on unrelated data characteristics.

3) *Fitness-function*: The choice of fitness function has a substantially larger impact on the quality of distribution tail augmentation than other design factors. Figure 3 presents a comparative analysis of different fitness functions and the resulting evaluation metrics obtained after optimization. Specifically, we consider the Jensen–Shannon divergence, the Kolmogorov–Smirnov test statistic, the Wasserstein distance, and the mean squared error between the eigenvectors of mutual information matrices as alternative fitness objectives.

Using the Kolmogorov–Smirnov test as the fitness function yields the highest values across both downstream machine learning performance metrics and statistical similarity metrics. This result indicates that directly optimizing for distributional alignment in terms of cumulative distribution functions is particularly effective for improving predictive utility. However, this choice is associated with nearly the worst performance in privacy-related metrics, with only the Wasserstein-based objective exhibiting higher privacy risk. This trade-off suggests

Table II
COMPREHENSIVE EVALUATION OF SYNTHETIC DATA GENERATION MODELS: DATA CURATION. **BOLD** VALUES INDICATE BEST PERFORMANCE WITHIN EACH METRIC CATEGORY.

Metric	CTGAN		DDPM		DPGAN	
	pre	post	pre	post	pre	post
Jensen–Shannon div. ↓	0.03	0.02	0.08	0.08	0.05	0.05
Inv. Kullback-Leibler div. ↑	0.49	0.44	0.09	0.09	0.16	0.15
Kolmogorov-Smirnov test ↑	0.72	0.76	0.39	0.40	0.37	0.36
Linear syn id ↑	0.92	0.99	0.84	0.97	0.78	0.99
Linear syn ood ↑	0.88	0.99	0.83	0.97	0.77	0.99
Xgb syn id ↑	0.92	0.98	0.56	0.64	0.82	0.88
Xgb syn ood ↑	0.93	0.98	0.55	0.63	0.90	0.88
Detection xgb	0.97	0.93	0.99	1.00	0.99	0.98
Detection linear	0.92	0.72	0.96	0.74	1.00	0.89
Delta-presence ↓	1.59	1.47	1.77	2.16	18.50	10.50
Identifiability ↓	0.34	0.32	0.03	0.03	0.08	0.07

Table III
COMPREHENSIVE EVALUATION OF SYNTHETIC DATA GENERATION MODELS: TAIL EXTENSION. **BOLD** VALUES INDICATE BEST PERFORMANCE WITHIN EACH METRIC CATEGORY.

Metric	CTGAN		DDPM		DPGAN		QWEN	
	pre	post	pre	post	pre	post	pre	post
Jensen–Shannon div. ↓	0.03	0.03	0.08	0.06	0.05	0.05	0.02	0.02
Inv. Kullback-Leibler div. ↑	0.36	0.46	0.11	0.11	0.14	0.14	0.65	0.71
Kolmogorov-Smirnov test ↑	0.69	0.74	0.46	0.60	0.37	0.38	0.81	0.84
Linear syn id ↑	0.90	0.93	0.71	0.79	0.58	0.69	0.99	1.00
Linear syn ood ↑	0.93	0.94	0.70	0.77	0.55	0.69	0.99	1.00
Xgb syn id ↑	0.90	0.86	0.62	0.70	0.69	0.84	1.00	0.99
Xgb syn ood ↑	0.91	0.86	0.59	0.70	0.88	0.84	0.99	0.99
Detection xgb	0.95	0.73	0.99	1.00	0.98	0.99	0.70	0.71
Detection linear	0.89	0.73	0.92	0.50	1.00	1.00	0.48	0.55
Delta-presence ↓	1.45	1.05	1.11	1.18	20.00	20.00	1.00	1.11
Identifiability ↓	0.50	0.58	0.06	0.11	0.17	0.17	0.47	0.58

that aggressively enforcing distributional similarity in the tails may amplify memorization effects or reduce variability in sensitive regions of the data.

In contrast, the Jensen–Shannon divergence emerges as the most balanced objective. While it does not achieve the absolute maximum in machine learning performance, it provides consistently strong results across predictive metrics, maintains high statistical fidelity, and simultaneously yields low values for privacy risk metrics. This behavior indicates that the Jensen–Shannon divergence offers a smoother and more globally constrained optimization landscape, mitigating overfitting to extreme regions while preserving essential structural properties of the data.

B. Full pipeline

We next compare the results obtained before and after applying the full pipeline implemented within the proposed

framework. Overall, the pipeline demonstrates a consistent ability to preserve or improve the quality of synthetic data while simultaneously reducing privacy risks across different generative models.

For the CTGAN-based pipeline, we observe that the quality of the synthetic data is maintained or slightly improved across the majority of evaluation metrics. At the same time, there is a notable reduction in privacy leakage risk: the Detection XGB score decreases, and the Delta Presence metric is reduced by nearly a factor of two, while identifiability remains at a comparable level. This indicates that the pipeline effectively mitigates membership and record inference risks without sacrificing downstream utility.

A similar trend is observed for the DDPM-based pipeline. The application of the full pipeline does not degrade synthetic data quality: distributional similarity metrics and classifier per-

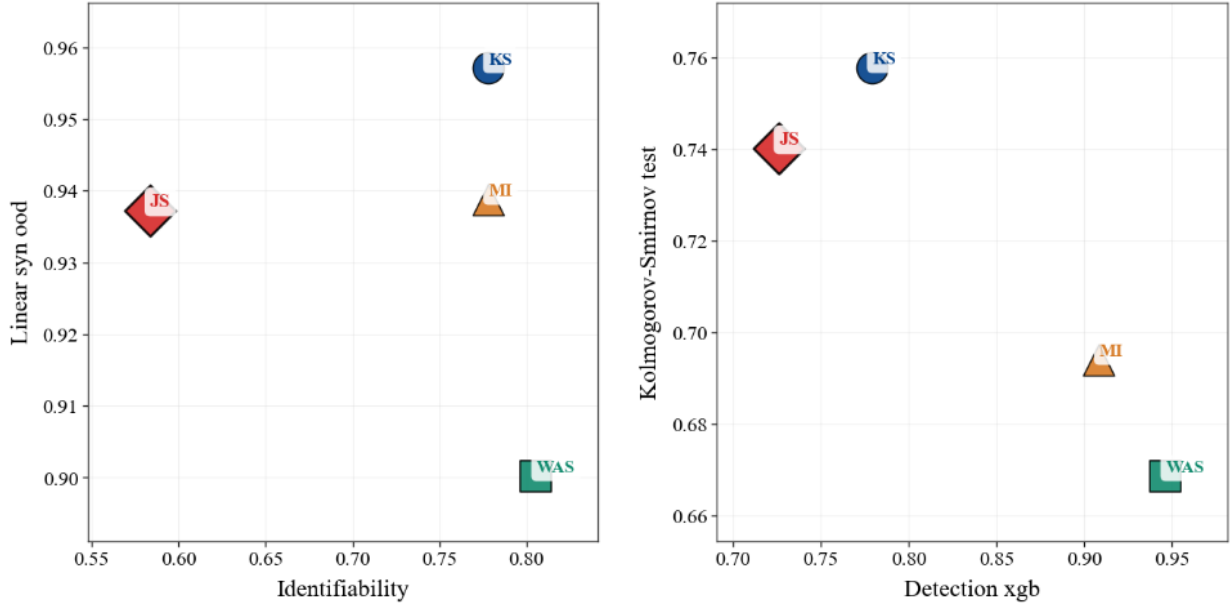


Figure 3. Schematic representation of the synthetic data curation procedure, showing iterative selection and refinement of data subsets to enhance predictive model outcomes.

Table IV
COMPREHENSIVE EVALUATION OF SYNTHETIC DATA GENERATION MODELS. **BOLD** VALUES INDICATE BEST PERFORMANCE WITHIN EACH METRIC CATEGORY.

Metric	CTGAN		DDPM		DPGAN		GPT-OSS	
	Only gen	Full	Only gen	Full	Only gen	Full	Only gen	Full
Nearest Syn Neighbor ↓	0.09	0.20	0.21	0.24	0.42	0.28	0.15	0.20
Close Values Probability ↑	0.87	0.67	0.68	0.56	0.30	0.33	0.68	0.66
Distant Values Probability ↓	0.02	0.02	0.09	0.03	0.14	0.03	0.01	0.02
Jensen-Shannon Distance ↓	0.02	0.01	0.04	0.03	0.07	0.04	0.02	0.02
Inverse KL Divergence ↑	0.73	0.80	0.54	0.57	0.32	0.51	0.78	0.77
Kolmogorov-Smirnov Test ↑	0.82	0.86	0.70	0.75	0.37	0.58	0.85	0.84
Linear syn id ↑	0.86	0.89	0.69	0.92	0.68	0.64	0.81	0.82
Linear syn ood ↑	0.84	0.89	0.77	0.93	0.65	0.66	0.82	0.81
MLP syn id ↑	0.52	0.53	0.50	0.51	0.50	0.62	0.46	0.52
MLP syn ood ↑	0.52	0.53	0.50	0.49	0.51	0.60	0.46	0.52
XGB syn id ↑	0.93	0.93	0.76	0.81	0.61	0.68	0.90	0.91
XGB syn ood ↑	0.92	0.92	0.74	0.83	0.47	0.70	0.91	0.91
Detection XGB ↓	0.76	0.79	0.74	0.89	1.00	0.98	0.71	0.87
Delta Presence ↓	5.94	3.09	1.32	1.48	6.44	52.91	1.97	2.90
Identifiability Score ↓	0.24	0.29	0.19	0.26	0.03	0.06	0.26	0.26

formance metrics (Linear/MLP/XGB for both in-distribution and out-of-distribution synthetic data) are either preserved or exhibit slight improvements. Concurrently, both Detection XGB and Delta Presence decrease, suggesting that the pipeline enhances the safety of the generation process and provides stronger protection of the original records while retaining their analytical value.

For DPGAN, the full pipeline leads to a pronounced improvement in data quality across most metrics. In particular, we observe reductions in the Nearest Synthetic Neighbor distance, Jensen–Shannon divergence, and Kullback–Leibler

divergence, as well as a decrease in the probability of extreme outlier values. Together, these effects indicate a more realistic and well-calibrated synthetic data distribution, mitigating common artifacts associated with differentially private generative models.

In contrast, for GPT-OSS, the addition of the pipeline results in minimal changes to distributional similarity metrics and classifier performance. Improvements in Linear/MLP/XGB accuracy are limited to only a few percentage points. This suggests that, for large pretrained generative models with strong implicit regularization, the marginal benefits of the

proposed pipeline are more constrained compared to classical generative architectures.

Taken together, these results demonstrate that the proposed pipeline consistently improves the privacy–utility trade-off for a wide range of generative models, with particularly strong gains for models that are more sensitive to data sparsity and distributional artifacts.

C. Hyperparameter optimization of task-specific models

D. Ablation Study

VI. DISCUSSION AND LIMITATIONS

A. Implications of Data Curation for Scientific Data Creation

B. Limitation of Predictive ML models Training on Synthetic Data

C. Limitation of LLM Training on Synthetic Data

D. Ethical and Regulatory Considerations in Synthetic Data

Synthetic data offers valuable opportunities for research and model development but also carries ethical and regulatory risks. Misuse—such as producing biased records or unbalanced demographic profiles—can amplify disparities and lead to flawed conclusions. Even synthetic datasets may encode patterns from real individuals, raising privacy and re-identification concerns. Compliance with frameworks like HIPAA or GDPR therefore remains essential.

Responsible use requires transparency, documentation of generation methods, and careful curation. In this study, only publicly available open-source datasets were used, reducing risks of sensitive data exposure. In practical applications, however, raw data must be rigorously anonymized before training generative models. To address these challenges, our work focuses on the privacy–utility trade-off, ensuring that synthetic data preserves research value while maintaining ethical and legal safeguards.

VII. CONCLUSION

VIII. APPENDIX

All code and generation results are available in repository

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¹<https://github.com/Samp0rt/TabAutoSyn>

Table V

COMPREHENSIVE EVALUATION OF SYNTHETIC DATA GENERATION MODELS: HYPERPARAMETER OPTIMIZATION OF DPGAN. **BOLD** VALUES INDICATE BEST PERFORMANCE WITHIN EACH METRIC CATEGORY.

Metric	OLD CONFIG	KS_TEST + XGB	KS_TEST + MLP	INV_KL + XGB	INV_KL + MLP
Nearest Syn Neighbor ↓	0.19	0.24	0.21	0.32	0.23
Close Values Probability ↑	0.61	0.54	0.62	0.45	0.53
Distant Values Probability ↓	0.01	0.02	0.02	0.08	0.01
Jensen-Shannon Distance ↓	0.07	0.05	0.06	0.06	0.07
Inverse KL Divergence ↑	0.37	0.55	0.41	0.52	0.41
Kolmogorov-Smirnov Test ↑	0.45	0.57	0.46	0.45	0.37
Linear syn id ↑	0.68	0.38	0.39	0.52	0.46
Linear syn ood ↑	0.69	0.41	0.40	0.55	0.48
MLP syn id ↑	0.50	0.51	0.38	0.45	0.53
MLP syn ood ↑	0.51	0.52	0.38	0.44	0.53
XGB syn id ↑	0.50	0.49	0.32	0.52	0.52
XGB syn ood ↑	0.39	0.41	0.40	0.51	0.56
Detection XGB ↓	0.96	0.99	1.00	0.99	0.98
Delta Presence ↓	44.30	35.40	6.53	63.07	223.12
Identifiability Score ↓	0.08	0.03	0.08	0.02	0.02

Table VI

COMPREHENSIVE EVALUATION OF SYNTHETIC DATA GENERATION MODELS: HYPERPARAMETER OPTIMIZATION OF DDPM. **BOLD** VALUES INDICATE BEST PERFORMANCE WITHIN EACH METRIC CATEGORY.

Metric	OLD CONFIG	KS_TEST + XGB	KS_TEST + MLP	INV_KL + XGB	INV_KL + MLP
Nearest Syn Neighbor ↓	0.18	0.15	0.26	0.24	0.16
Close Values Probability ↑	0.69	0.77	0.58	0.60	0.79
Distant Values Probability ↓	0.03	0.03	0.09	0.09	0.09
Jensen-Shannon Distance ↓	0.03	0.03	0.05	0.05	0.04
Inverse KL Divergence ↑	0.60	0.59	0.42	0.43	0.61
Kolmogorov-Smirnov Test ↑	0.77	0.74	0.64	0.62	0.69
Linear syn id ↑	0.63	0.60	0.50	0.45	0.65
Linear syn ood ↑	0.62	0.62	0.53	0.48	0.62
MLP syn id ↑	0.53	0.53	0.57	0.50	0.63
MLP syn ood ↑	0.53	0.53	0.58	0.50	0.60
XGB syn id ↑	0.80	0.80	0.60	0.60	0.89
XGB syn ood ↑	0.81	0.82	0.60	0.62	0.90
Detection XGB ↓	0.66	0.68	0.81	0.78	0.69
Delta Presence ↓	1.54	1.53	231.92	6.50	3.95
Identifiability Score ↓	0.26	0.32	0.02	0.10	0.27

Table VII

COMPREHENSIVE EVALUATION OF SYNTHETIC DATA GENERATION MODELS: HYPERPARAMETER OPTIMIZATION OF CTGAN. **BOLD** VALUES INDICATE BEST PERFORMANCE WITHIN EACH METRIC CATEGORY.

Metric	OLD CONFIG	KS_TEST + XGB	KS_TEST + MLP	INV_KL + XGB	INV_KL + MLP
Nearest Syn Neighbor ↓	0.15	0.07	0.10	0.07	0.16
Close Values Probability ↑	0.71	0.93	0.89	0.93	0.73
Distant Values Probability ↓	0.01	0.01	0.01	0.02	0.02
Jensen-Shannon Distance ↓	0.01	0.01	0.02	0.01	0.01
Inverse KL Divergence ↑	0.77	0.77	0.74	0.77	0.82
Kolmogorov-Smirnov Test ↑	0.87	0.85	0.85	0.86	0.85
Linear syn id ↑	0.66	0.81	0.82	0.81	0.71
Linear syn ood ↑	0.65	0.81	0.81	0.82	0.70
MLP syn id ↑	0.46	0.53	0.57	0.45	0.50
MLP syn ood ↑	0.46	0.53	0.59	0.45	0.51
XGB syn id ↑	0.73	0.86	0.79	0.88	0.73
XGB syn ood ↑	0.77	0.89	0.75	0.88	0.71
Detection XGB ↓	0.70	0.65	0.71	0.74	0.71
Delta Presence ↓	1.64	1.99	4.65	2.45	5.84
Identifiability Score ↓	0.31	0.28	0.33	0.33	0.23

Table VIII
COMPREHENSIVE EVALUATION OF SYNTHETIC DATA GENERATION MODELS: ABLATION STUDY OF CTGAN. **BOLD** VALUES INDICATE BEST PERFORMANCE WITHIN EACH METRIC CATEGORY.

Metric	Only generator	No hyperparam. optim.	No outliers extension	No curation	FULL PIPELINE
Nearest Syn Neighbor ↓	0.09	0.18	0.18	0.15	0.20
Close Values Probability ↑	0.87	0.67	0.68	0.76	0.67
Distant Values Probability ↓	0.02	0.02	0.02	0.02	0.02
Jensen-Shannon Distance ↓	0.02	0.02	0.01	0.01	0.01
Inverse KL Divergence ↑	0.73	0.73	0.77	0.80	0.80
Kolmogorov-Smirnov Test ↑	0.82	0.82	0.86	0.84	0.86
Linear syn id ↑	0.86	0.87	0.86	0.85	0.89
Linear syn ood ↑	0.84	0.87	0.86	0.84	0.89
MLP syn id ↑	0.52	0.61	0.53	0.51	0.53
MLP syn ood ↑	0.52	0.61	0.53	0.52	0.53
XGB syn id ↑	0.93	0.94	0.94	0.92	0.93
XGB syn ood ↑	0.92	0.92	0.91	0.92	0.92
Detection XGB ↓	0.76	0.96	0.84	0.70	0.79
Delta Presence ↓	5.94	2.64	2.23	1.87	3.09
Identifiability Score ↓	0.24	0.25	0.27	0.27	0.29

Table IX
COMPREHENSIVE EVALUATION OF SYNTHETIC DATA GENERATION MODELS: ABLATION STUDY OF DDPM. **BOLD** VALUES INDICATE BEST PERFORMANCE WITHIN EACH METRIC CATEGORY.

Metric	Only generator	No hyperparam. optim.	No outliers extension	No curation	FULL PIPELINE
Nearest Syn Neighbor ↓	0.21	0.28	0.24	0.14	0.24
Close Values Probability ↑	0.68	0.55	0.56	0.80	0.56
Distant Values Probability ↓	0.09	0.08	0.03	0.03	0.03
Jensen-Shannon Distance ↓	0.04	0.04	0.03	0.03	0.03
Inverse KL Divergence ↑	0.54	0.54	0.58	0.62	0.57
Kolmogorov-Smirnov Test ↑	0.70	0.73	0.76	0.77	0.75
Linear syn id ↑	0.69	0.86	0.91	0.84	0.92
Linear syn ood ↑	0.77	0.86	0.92	0.84	0.93
MLP syn id ↑	0.50	0.61	0.50	0.61	0.51
MLP syn ood ↑	0.50	0.63	0.50	0.61	0.49
XGB syn id ↑	0.76	0.78	0.92	0.81	0.81
XGB syn ood ↑	0.74	0.83	0.91	0.82	0.83
Detection XGB ↓	0.74	0.93	0.82	0.67	0.89
Delta Presence ↓	1.32	1.46	1.50	1.43	1.48
Identifiability Score ↓	0.19	0.14	0.27	0.32	0.26

Table X
COMPREHENSIVE EVALUATION OF SYNTHETIC DATA GENERATION MODELS: ABLATION STUDY OF DPGAN. **BOLD** VALUES INDICATE BEST PERFORMANCE WITHIN EACH METRIC CATEGORY.

Metric	Only generator	No hyperparam. optim.	No outliers extension	No curation	FULL PIPELINE
Nearest Syn Neighbor ↓	0.42	0.39	0.28	0.29	0.28
Close Values Probability ↑	0.30	0.29	0.36	0.34	0.33
Distant Values Probability ↓	0.14	0.08	0.04	0.03	0.03
Jensen-Shannon Distance ↓	0.07	0.06	0.05	0.04	0.04
Inverse KL Divergence ↑	0.32	0.34	0.47	0.56	0.51
Kolmogorov-Smirnov Test ↑	0.37	0.39	0.55	0.59	0.58
Linear syn id ↑	0.68	0.63	0.70	0.58	0.64
Linear syn ood ↑	0.65	0.64	0.71	0.59	0.66
MLP syn id ↑	0.50	0.51	0.62	0.52	0.62
MLP syn ood ↑	0.51	0.50	0.60	0.52	0.60
XGB syn id ↑	0.61	0.66	0.71	0.67	0.68
XGB syn ood ↑	0.47	0.63	0.67	0.64	0.70
Detection XGB ↓	1.00	0.99	0.99	0.95	0.98
Delta Presence ↓	6.44	8.34	143.68	82.84	52.91
Identifiability Score ↓	0.03	0.03	0.08	0.08	0.06

Table XI
 COMPREHENSIVE EVALUATION OF SYNTHETIC DATA GENERATION MODELS: ABLATION STUDY OF GPT-OSS. **BOLD** VALUES INDICATE BEST PERFORMANCE WITHIN EACH METRIC CATEGORY.

Metric	Only generator	No hyperparam. optim.	No outliers extension	No curation	FULL PIPELINE
Nearest Syn Neighbor ↓	0.15	-	0.19	0.16	0.20
Close Values Probability ↑	0.68	-	0.69	0.68	0.66
Distant Values Probability ↓	0.01	-	0.02	0.01	0.02
Jensen-Shannon Distance ↓	0.02	-	0.02	0.02	0.02
Inverse KL Divergence ↑	0.78	-	0.74	0.79	0.77
Kolmogorov-Smirnov Test ↑	0.85	-	0.83	0.84	0.84
Linear syn id ↑	0.81	-	0.81	0.82	0.82
Linear syn ood ↑	0.82	-	0.81	0.82	0.81
MLP syn id ↑	0.46	-	0.49	0.52	0.52
MLP syn ood ↑	0.46	-	0.48	0.52	0.52
XGB syn id ↑	0.90	-	0.88	0.91	0.91
XGB syn ood ↑	0.91	-	0.90	0.92	0.91
Detection XGB ↓	0.71	-	0.87	0.70	0.87
Delta Presence ↓	1.97	-	2.31	2.00	2.90
Identifiability Score ↓	0.26	-	0.29	0.27	0.26